

EE5203 L10 Study Sheet

UART and PC communication - Basri Erdogan

Essential question: Two devices share one wire and no clock. How do they agree on where each bit begins — and what happens when they disagree?

What you should be able to do

- Draw an 8N1 frame and count its bit times.
- Convert between baud, microseconds per byte, and lines per second.
- Compute BRR from PCLK and a baud rate, and the resulting error.
- Transmit formatted telemetry from the tick without blocking the loop.
- Receive by interrupt, re-arm correctly, and act in `main()`.

The frame

```
idle  START  D0 D1 D2 D3 D4 D5 D6 D7  STOP  idle
high  low    <---- LSB first ----->  high
      |<----- 10 bit times ----->|
```

8N1 = 8 data bits, No parity, 1 stop bit. One byte therefore costs **10 bit times** — the framing overhead is 20 %.

$$\text{time per byte} = \frac{10}{\text{baud}} \quad \text{bytes per second} = \frac{\text{baud}}{10}$$

Baud	µs/byte	bytes/s	18-char line
9 600	1041.67	960	18.75 ms
57 600	173.61	5 760	3.13 ms
115 200	86.81	11 520	1.56 ms

There is no clock wire. Both ends agree the rate in advance; the receiver re-synchronises on each start bit and then samples in the middle of each bit time. Agreement on baud, data bits, parity, and stop bits must be exact — a UART cannot report a disagreement, only hand you its consequences.

Vocabulary table

Name	Job
Baud	bits per second on the wire , including framing
BRR	the divider that makes the baud from PCLK
TXE	transmit register empty — write the <i>next</i> byte
TC	transmission complete — the line is genuinely idle
RXNE	a byte has arrived; reading DR clears it
ORE	overrun — a byte arrived before you read the last one; it is lost
VCP	virtual COM port — the ST-LINK's USB-to-serial bridge

Computing BRR

USART2 is on **APB1**; at the default HSI, PCLK1 = 16 MHz. Oversampling by 16:

$$\text{USARTDIV} = \frac{f_{\text{PCLK}}}{16 \times \text{baud}}$$

BRR stores this as fixed point: **top 12 bits = integer part, bottom 4 bits = sixteenths.**

For 115 200: $16\,000\,000 / (16 \times 115\,200) = 8.6806 \rightarrow$ mantissa **8**, fraction $0.6806 \times 16 = 11 \rightarrow \text{BRR} = (8 \ll 4) | 11 = \mathbf{139} = 0x08B \rightarrow \text{actual} = 16\,000\,000 / (16 \times 8.6875) = \mathbf{115\,108\,baud}$, error **−0.08 %**

Baud	BRR	Actual	Error
9 600	0x683	9 598	−0.02 %
38 400	0x1A1	38 369	−0.08 %
115 200	0x08B	115 108	−0.08 %
230 400	0x045	231 884	+0.64 %

A UART tolerates roughly $\pm 2\%$ total across both ends. Small errors are invisible; 0.64 % consumes a third of the budget before the other device has contributed anything.

The board already has the wire

PA2 (USART2_TX) and PA3 (USART2_RX) are wired to the ST-LINK, which presents a serial port over the same USB cable that flashes the board. No dongle, no level shifter. **Close the Serial Monitor before flashing** — an open port blocks the programmer.

Transmitting

```
char buf[48];
int n = snprintf(buf, sizeof buf, "%lu,%u\r\n",
                 (unsigned long)g_ms, (unsigned)raw);
HAL_UART_Transmit(&huart2, (uint8_t *)buf, (uint16_t)n, 100);
```

- **snprintf, never sprintf.** snprintf truncates at the buffer size; sprintf writes past the end and corrupts whatever is next on the stack.
- **HAL_UART_Transmit blocks** until the last byte is handed over — 1.56 ms for an 18-character line at 115 200. Acceptable in main() at 10 Hz; never acceptable in a callback.
- **\r\n**, not **\n** alone — most monitors need both to return to column 1.
- Comma-separated with a timestamp pastes straight into a spreadsheet.

TXE versus TC. TXE means “the holding register is free, give me the next byte” — the previous byte may still be shifting out. TC means the shift register is empty too and the line is idle. Powering down, or turning a half-duplex transceiver around, needs TC.

Receiving

```
HAL_UART_Receive_IT(&huart2, &rx_byte, 1); /* arm once */

void HAL_UART_RxCpltCallback(UART_HandleTypeDef *huart)
{
    if (huart->Instance == USART2) {
        rx_ring[head] = rx_byte;
        head = (head + 1) % RX_SIZE;
        HAL_UART_Receive_IT(&huart2, &rx_byte, 1); /* RE-ARM - mandatory */
    }
}
```

- **Never HAL_UART_Receive blocking.** It waits for a human; meanwhile nothing else in the system runs.
- **The re-arm is mandatory.** HAL disarms after each completed receive. Omit it and you receive exactly one byte, ever — a silence that looks like broken hardware.
- The callback **records**; main() **decides**. Same discipline as every interrupt since L06.
- Reading DR clears RXNE — read-to-clear, like the ADC’s EOC.

ORE. Set when a byte arrives before the previous one was read. That byte is gone, and HAL delivers nothing further until the error is cleared. Usual causes: blocking somewhere long, a missing re-arm, or a breakpoint held while data arrives.

SFR evidence

Claim	Evidence
the baud divider is right	USART2->BRR = 0x08B for 115 200 at 16 MHz
the pins belong to the USART	MODER = 10 for PA2/PA3; AFR[0] selects AF7

Claim	Evidence
a byte is waiting	USART2->SR RXNE = 1; it clears when DR is read
bytes were lost	USART2->SR ORE = 1 after a held breakpoint
the line is idle	TC = 1 after the last byte

Common mistakes

- Baud mismatch between board and monitor — consistent garbage, no warning.
- Receive interrupt never re-armed — one byte, then permanent silence.
- `sprintf` into a small buffer — silent corruption.
- `HAL_UART_Transmit` inside a callback — milliseconds of blocking in interrupt context.
- `ORE` never cleared — reception dead for good.
- `\n` without `\r` — the display stair-steps.
- Serial Monitor left open while flashing — the programmer cannot claim the port.
- `printf` with floats for telemetry — huge, slow, one transmit per character.

Mastery self-check

- ☐ I can draw an 8N1 frame and say why a byte costs 10 bit times.
- ☐ I can compute $\mu\text{s}/\text{byte}$ and max lines/s for any baud.
- ☐ I can compute `BRR` and the resulting baud error.
- ☐ I can state the $\pm 2\%$ tolerance and why it matters.
- ☐ I can distinguish `TXE` from `TC`, and `RXNE` from `ORE`.
- ☐ I can write an interrupt receive with a correct re-arm.
- ☐ I can diagnose garbage, silence, and one-byte-then-nothing by symptom.

Five review questions

1. At 57 600 baud, 8N1, how long does a 17-character line take? What fraction of a 50 ms budget is that, and how many such lines per second could the link carry flat out?
2. Compute `BRR` for 19 200 baud with `PCLK1` = 16 MHz. Give mantissa, fraction, hex value, actual baud, and error.
3. A student's board sends its banner correctly, but every character the *PC* sends is ignored after the first one. Name the single missing line.
4. Explain why `HAL_UART_Transmit(&huart2, buf, 18, 100)` is acceptable in `main()` at 10 Hz but not inside `HAL_TIM_PeriodElapsedCallback` on a 1 ms tick. Use numbers.
5. Your monitor shows readable text that occasionally drops characters, and `ORE` is set. Give two distinct causes and the fix for each.

See the L10 worksheet for extended practice. Worked solutions are released after the practice window.